



Common Ports & Protocols

Use this when: you read a firewall rule, scan or netstat output and need the service behind a port.

CCNA · Network+ · A+ Core 1
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Port map: 32 ports to know

Port · protocol · transport

Remote & files	20/21 FTP TCP	22 SSH TCP	23 Telnet TCP	69 TFTP UDP	137-139 NetBIOS UDP/TCP	445 SMB/CIFS TCP	3389 RDP TCP/UDP	5900 VNC TCP
Web & email	80 HTTP TCP	443 HTTPS TCP/UDP	25 SMTP TCP	587 Mail submit TCP	110 POP3 TCP	995 POP3S TCP	143 IMAP TCP	993 IMAPS TCP
Network services	53 DNS UDP/TCP	67/68 DHCP UDP	123 NTP UDP	161/162 SNMP UDP	514 Syslog UDP	6514 Syslog-TLS TCP	5060 SIP UDP/TCP	5061 SIP-TLS TCP
Identity & data	88 Kerberos UDP/TCP	389 LDAP TCP/UDP	636 LDAPS TCP	49 TACACS+ TCP	1812/13 RADIUS UDP	1433 MS SQL TCP	3306 MySQL TCP	5432 PostgreSQL TCP

Encrypted (TLS / SSH) Cleartext (readable on the wire) Depends on config

Swap cleartext for secure

Telnet 23	→	SSH 22	encrypted
FTP 21 · TFTP 69	→	SFTP / SCP 22	over SSH
HTTP 80	→	HTTPS 443	TLS
POP3 110 · IMAP 143	→	995 · 993	implicit TLS
LDAP 389	→	LDAPS 636	or StartTLS on 389
SNMP v1/v2c	→	SNMPv3	adds auth + encryption

Port ranges

RFC 6335

0-1023	1024-49151	49152-65535
Well-known system services	Registered vendor apps	Dynamic client source ports
TCP	Reliable, handshake: web, SSH, mail, SMB	
UDP	Fast, no handshake: DNS, DHCP, NTP, VoIP	
Both	DNS: TCP 53 for zone transfers, big replies	

What is listening on this host?

<code>netstat -ano findstr LISTENING</code>	Windows
<code>Get-NetTCPConnection -State Listen</code>	PowerShell
<code>ss -tulpn</code>	Linux
<code>Test-NetConnection 192.0.2.10 -Port 22</code>	Remote port open?
<code>tasklist /FI "PID eq 4321"</code>	Who owns the PID?
<code>lsof -i :443</code>	Who uses 443?

CAUTION `ss -p` and `lsof` need `sudo` to see other users' processes. Exit the root shell when done.

Firewall rules of thumb

DANGER Never open these to the internet:

23 Telnet **135-139** **445 SMB** **3389 RDP**

1433 **3306** **161 SNMP**

Default	Deny all inbound, allow only what a service needs
Remote	RDP/SSH through a VPN or gateway with MFA
Verify	A port is a hint, not proof: check the service
Log	Log denied traffic and review it weekly

REMEMBER

- **FTP:** 21 = control, 20 = data (active mode)
- **SNMP:** 161 = manager polls, 162 = traps
- **RADIUS** = UDP, **TACACS+** = TCP 49

- **DHCP:** server 67, client 68, both UDP
- **Mail:** 25 server-to-server, 587 client sends
- **SSH 22** also carries SFTP and SCP